ADVANCED MACHINE LEARNING · PROJECT 21 · PHASE 1

Day-Ahead Flight Delay Risk Prediction

Leakage-free features and calibrated classifiers for New York departures, 2013

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github.com/techatanu/Flight_Delay_Risk

We predict delay risk the night before departure

Days before d

Outcomes of all earlier days are known

Night of day d – 1

Decision time: score every flight of day d

Day d

Flight operates; the label is known after landing

Allowed at decision time

- Full published schedule of day d (carrier, route, times, aircraft)
- Outcomes of strictly earlier days

Not allowed

- Departure delay, actual times, air time
- Observed weather
- Any outcome from day d itself

Target and output

- Delayed = arrival delay ≥ 15 min (BTS definition)
- Output: calibrated P(delay) per flight
- Used for a what-if rescheduling analysis

Literature review

Delay prediction

Sternberg et al. (2017) review the field. Rebollo & Balakrishnan (2014): random forests lose accuracy as the prediction horizon grows.

Weather and deep models

Choi et al. (2016), Gui et al. (2020) and Yazdi et al. (2020) rely on weather, ADS-B or near-departure state, none of which is known a day ahead.

Delay propagation

Pyrgiotis et al. (2013): delays spread through aircraft rotations. This motivates our rotation and recent-delay features.

Models

Gradient boosting (Friedman, 2001) and XGBoost (Chen & Guestrin, 2016); logistic regression as the required baseline.

Calibration and shift

Platt (1999), isotonic regression (Zadrozny & Elkan, 2002), ECE (Guo et al., 2017). A changed base rate needs a prior correction (Saerens et al., 2002).

Evaluation and leakage

PR curves are more informative than ROC under imbalance (Saito & Rehmsmeier, 2015). Leakage formalised by Kaufman et al. (2012).

Our gap: strictly day-ahead information, checked by an automated test, a chronological split, and calibration under temporal shift.

Dataset: every NYC departure in 2013

Kaggle copy of nycflights13 (BTS on-time data), airports EWR, JFK and LGA

336,776
scheduled departures

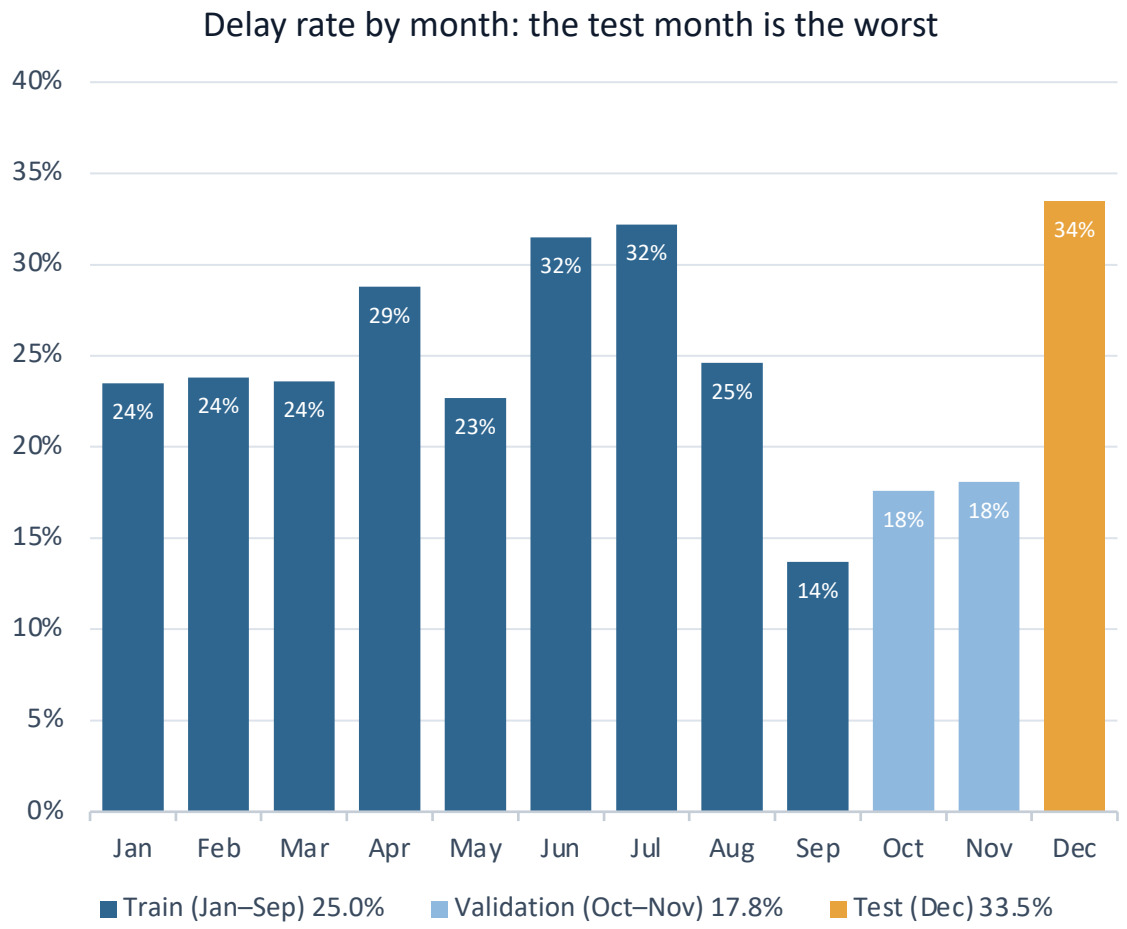
16 · 105
carriers · destinations

24.5%
labelled flights delayed ≥ 15 min

9,430
cancelled or diverted: no label, kept in volume counts

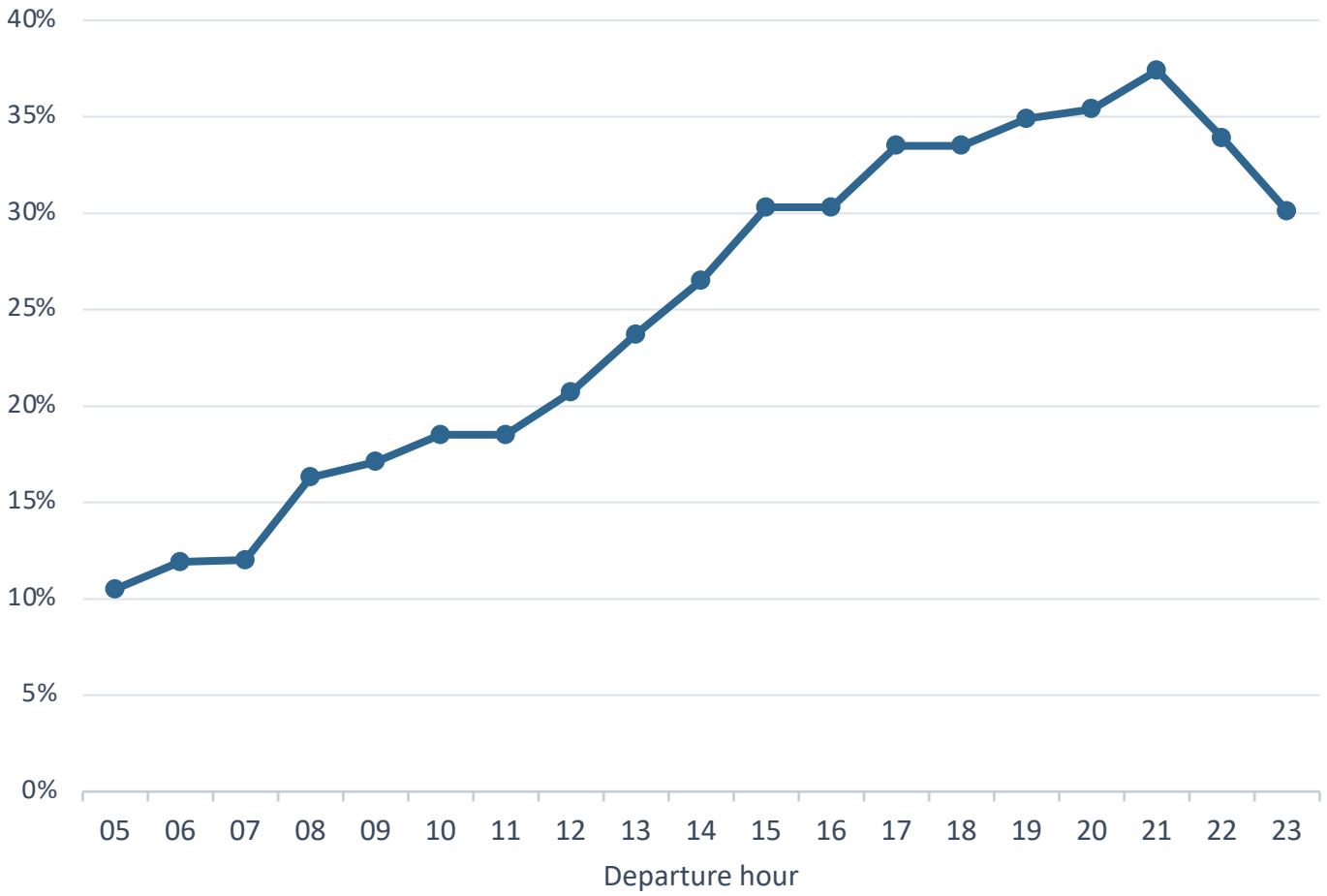
0.915
corr. with arrival delay: departure delay removed

2,512
rows without tail number; no duplicate ids

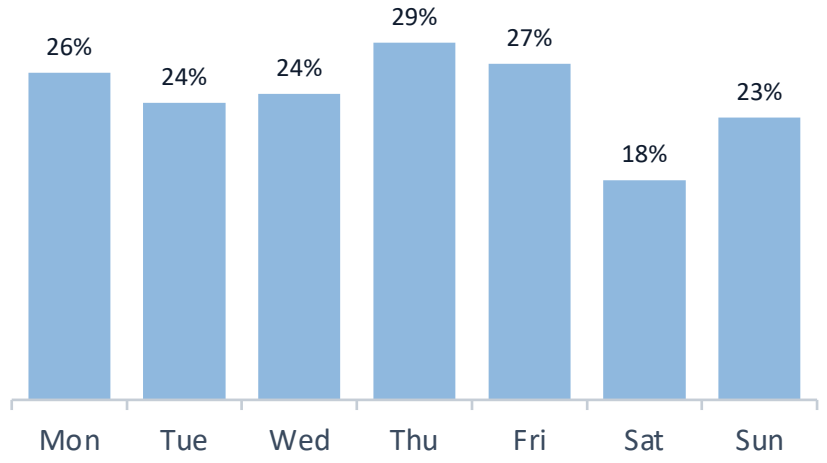


Delays build up through the day and carry over

Delay rate by scheduled departure hour



By day of week



0.40 lag-1 autocorrelation of the daily delay rate

Bad days spill into the next day, but the effect fades fast (0.13 at lag 2): yesterday's outcomes are useful features.

35 features, all known the night before

3 categorical (carrier, origin, destination) + 32 numeric, in five groups

Calendar

sin/cos of departure and arrival time and of weekday, weekend flag, days to nearest holiday

Schedule

departure hour, distance, overnight arrival flag

Congestion

scheduled departures per origin-day, origin-hour, destination-day, carrier-day; aircraft's legs that day

Long-run history

smoothed past delay rate per carrier, origin, destination, route, carrier-route, origin-hour, flight number, aircraft

Recent history

7-day rates per origin, carrier, destination; yesterday's system and origin delay rate, cancellations, mean delay

Past-only history

$$K_g(d) = \Sigma_{d' < d}\, k_g(d')$$

counts from earlier days only: cumulative sum, then a one-day shift

Smoothing (m = 20)

$$r_g = (K_g + m \cdot p_0) \,/\, (N_g + m)$$

Beta-prior mean: rare routes and flight numbers shrink to the system rate

Cyclic time

sin/cos of minute of day / 1440, so 23:59 and 00:00 are neighbours

Removed on purpose

day of year and month: Oct–Dec never appear in training

Honest evaluation: leakage test and time split



244,737 flights · 25.0% delayed

55,589 · 17.8%

27,020 · 33.5%

Automated leakage test (notebook 02)

1. Build all 35 features
2. Randomise every outcome on and after a cutoff date
3. Rebuild the features
4. Every feature up to the cutoff must be identical

Passed. A deliberately leaky version (no one-day shift) is caught.

Chosen on validation only

- LR: C = 10
- XGBoost: depth 4, min_child_weight 2000, λ = 20, 308 trees
- Platt calibration, 28-day rolling window
- Threshold: lowest with precision ≥ 0.5

December is scored once.

Models, calibration and metrics

Logistic regression

$$\min -\Sigma\,[y \log p + (1-y)\log(1-p)] + \|w\|^2 / 2C$$

- Convex: unique optimum; $\Sigma\, p_i = \Sigma\, y_i$ (calibrated in the large)
- One-hot categoricals, standardised numerics; C = 10

XGBoost

$$w^* = -G / (H + \lambda),\ g = p - y,\ h = p(1-p)$$

- Second-order boosting with leaf-weight penalty λ
- Depth 4, min_child_weight 2000, λ = 20, 308 trees

Calibration

$$q = \sigma(a \cdot \text{logit}\, p + c)$$

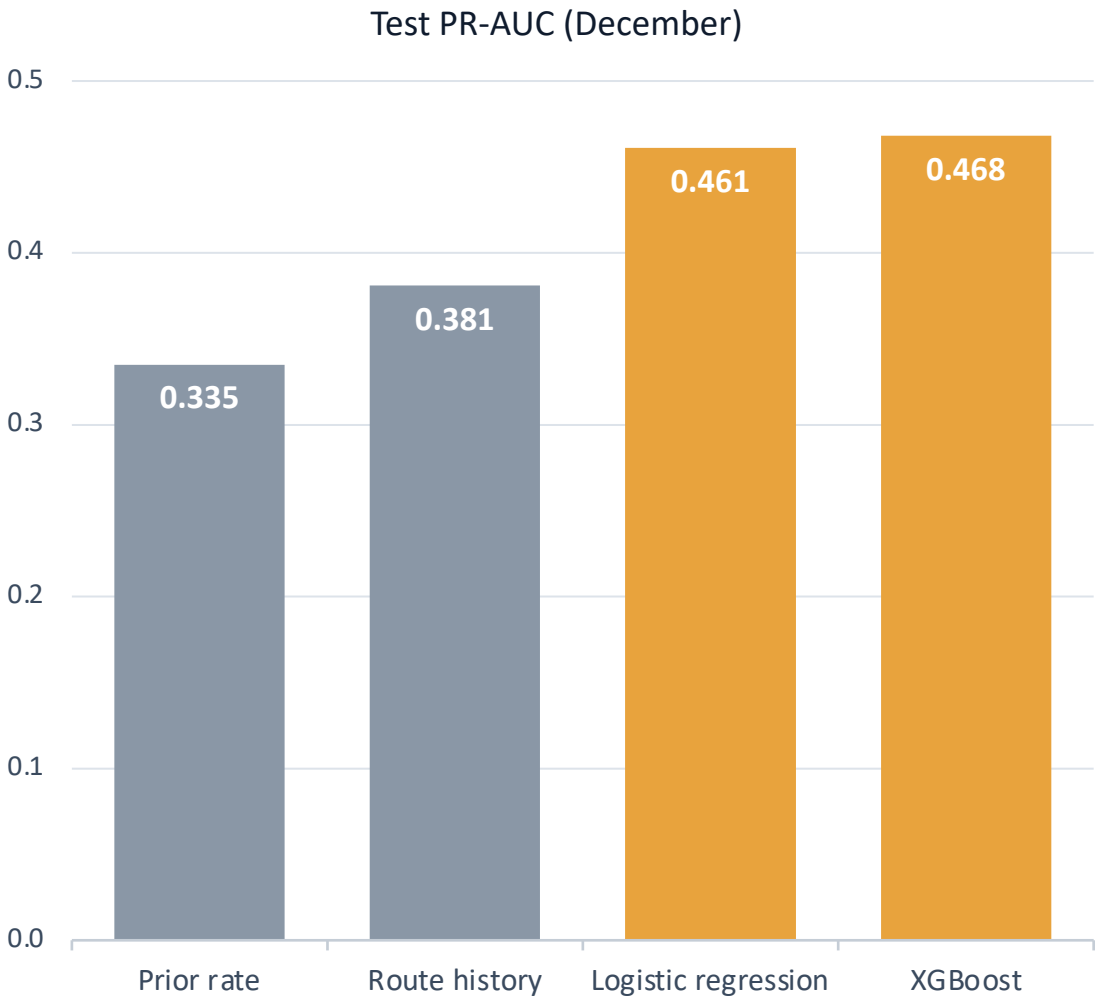
- Platt scaling beat isotonic on validation
- Rolling Platt: refit daily on the previous 28 days

Metrics

$$ECE = \Sigma \text{ over 10 bins of share} \times |\text{observed} - \text{predicted}|$$

- PR-AUC (headline), ROC-AUC, Brier score, ECE
- Operating point: recall at precision ≥ 0.5

Both models beat the baselines on December

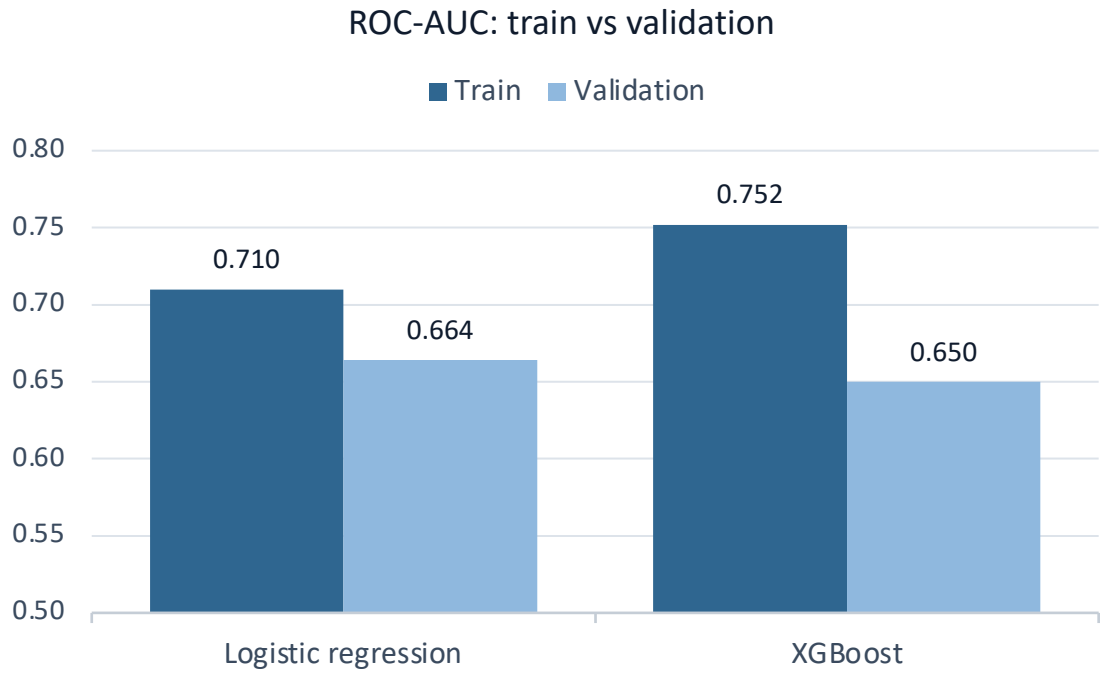


Model	PR-AUC	ROC-AUC	Brier	ECE
Prior rate	0.335	0.500	0.230	0.086
Route history	0.381	0.554	0.230	0.095
Logistic regression	0.461	0.640	0.222	0.103
XGBoost	**0.468**	**0.645**	**0.221**	**0.102**

Brier and ECE for LR and XGBoost after Platt calibration fitted on Oct–Nov

- On validation, LR is slightly ahead (PR-AUC 0.297 vs 0.276); on December XGBoost edges ahead
- Gains over the route baseline are clear but modest: without weather forecasts, day-level variance cannot be predicted
- ECE ≈ 0.10 for both: the calibration problem on the next slides

XGBoost needed very large leaves; class weights hurt



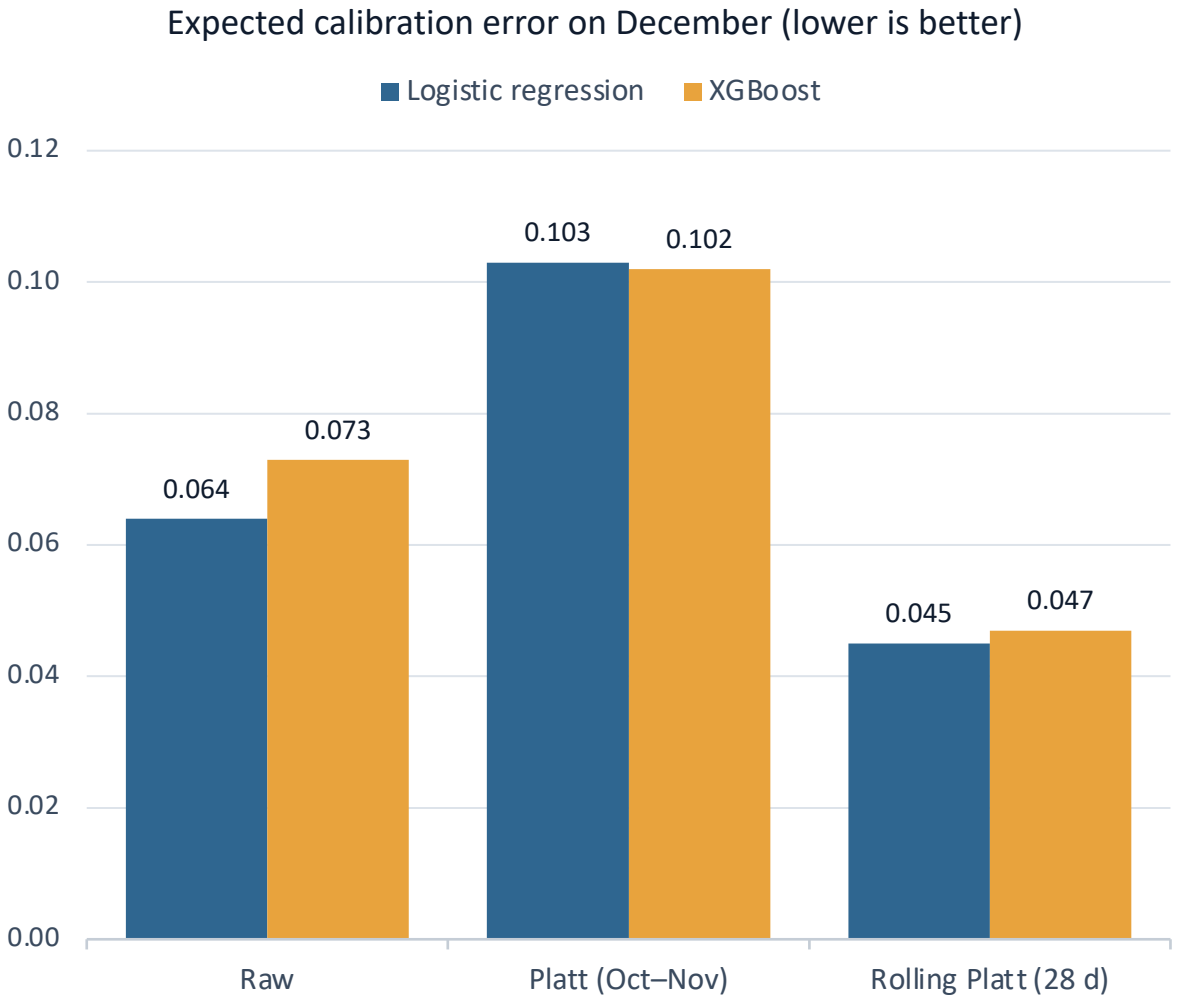
Day-level history features fingerprint individual training days, so small leaves memorise them. Validation picked min_child_weight = 2000: every leaf holds ≥ 8,000 flights (about nine days).

Class imbalance (raw, uncalibrated probabilities)

Model	Test PR-AUC	Test ECE	Mean P
LR, unweighted	**0.461**	**0.064**	**0.295**
LR, balanced weights	0.460	0.182	0.517
XGBoost, unweighted	**0.468**	**0.073**	**0.269**
XGBoost, pos. weight 3	0.466	0.153	0.488

- Weighting leaves PR-AUC unchanged but inflates probabilities (mean 0.5 vs 0.335 observed) and more than doubles ECE
- Kept unweighted; imbalance handled by a precision-based threshold. No SMOTE: synthetic flights would mix days and break the history features

Calibration breaks in December; rolling Platt fixes it



0.234 vs 0.335

XGBoost mean predicted vs observed December rate, Platt fitted on Oct–Nov

0.102 → 0.047

XGBoost ECE after refitting Platt daily on the previous 28 days

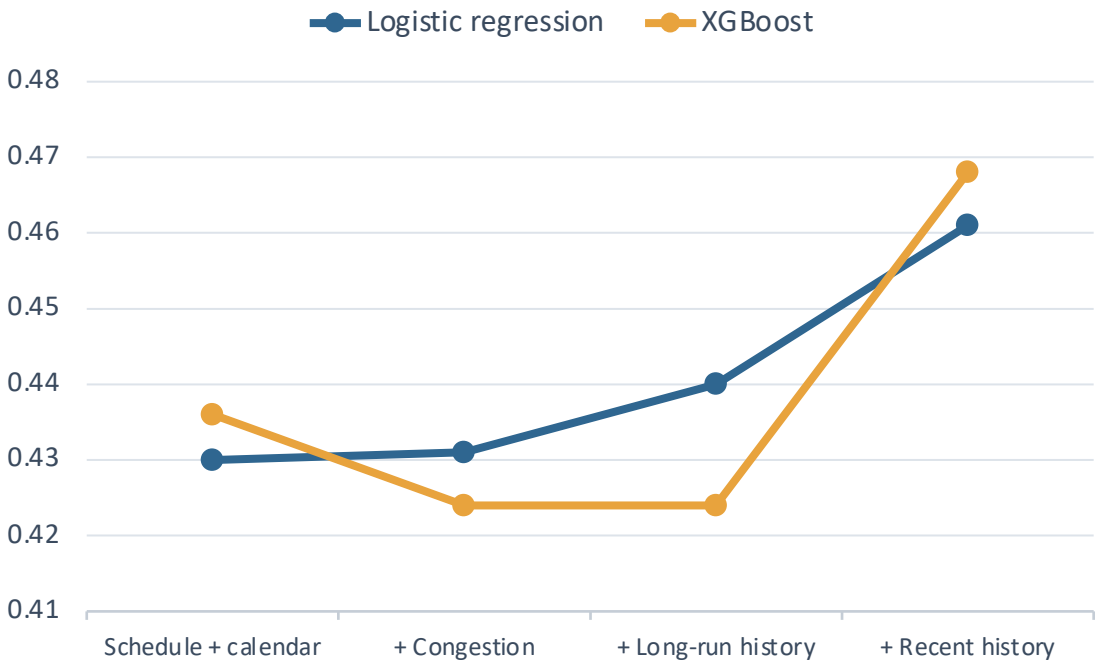
Operating point: precision ≥ 0.5

Model	Threshold	Precision	Recall
XGBoost	0.498	0.661	0.035
XGBoost (rolling cal.)	0.576	0.585	0.062

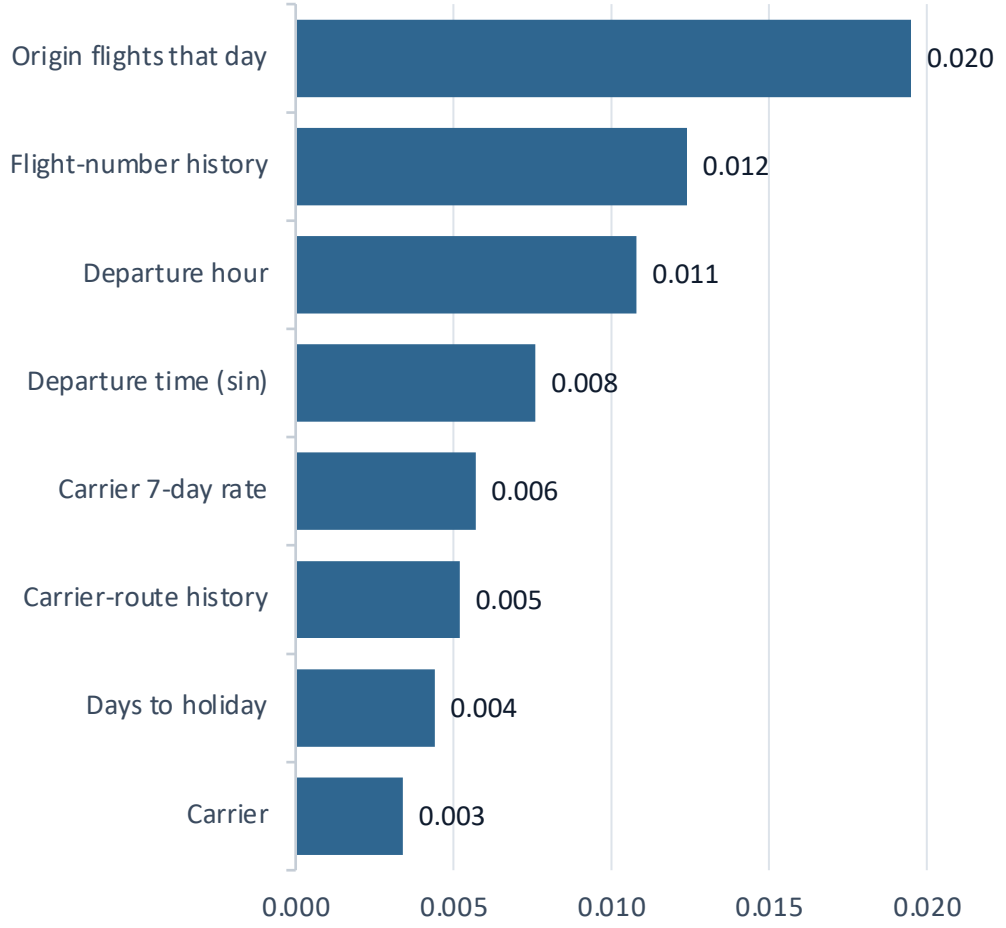
Precision holds but recall is low: few December flights pass a threshold set on calm months.

Recent history and congestion drive the predictions

Ablation: December PR-AUC as groups are added



XGBoost permutation importance (drop in PR-AUC)



Recent history gives the biggest December gain: disruption carries over from day to day. On calm validation months, congestion helps most.

What-if rescheduling: few flights benefit, mostly earlier

27,020
December test flights

→

1,979
risk ≥ 0.5 (52% truly delayed)

→

133
have a move cutting risk by ≥ 0.05

→

98
of those move earlier (35 later)

Rules and assumptions

- Try ±1 h and ±2 h on the same day
- Move if risk falls by ≥ 0.05; max 3 moves per origin-hour
- Moved flights: mean risk 0.571 → 0.510
- Expected delays among flagged: 1,123 → 1,115
- Associational only: ignores crews, rotations, connections

Sensitivity to the risk threshold

Risk threshold	Flagged	Moved	Earlier	Later
0.4	5,750	295	239	56
0.5	**1,979**	**133**	**98**	**35**
0.6	473	33	21	12

Much of the risk is attached to the day and route, not the hour, so ±2 h moves change little.

Team contributions

Atanu Adhikari
EDA and presentation

Data quality audit, delay target, seasonality and time-of-day analysis; built these slides

01_eda.ipynb · eda_month, eda_hour_dow, eda_daily, eda_carrier, eda_arr_delay figures · slides/

Sambhav Kumar
Feature engineering

35 past-only features with smoothing and cyclic encodings; automated leakage test; README

02_feature_engineering.ipynb · README.md

Nishant Ranjan Singh
Model training and report

LR and XGBoost tuning, overfitting, class imbalance and ablation; IEEE report

03_model_training.ipynb · tuning.csv, selected_models.json, ablation.csv, imbalance.csv · imbalance.tex · report/

Sankalp M Tellur
Evaluation and calibration

December test metrics, reliability and calibration, operating point, permutation importance

04_evaluation_calibration.ipynb · metrics, calibration, importance CSVs · 4 figures · 5 LaTeX tables

Prince Singh
Rescheduling and report

What-if rescheduling with explicit assumptions and threshold sensitivity; IEEE report

05_rescheduling.ipynb · reschedule_summary.csv · reschedule.png · 2 LaTeX tables · report/

Conclusion

- Leakage-free day-ahead pipeline, verified by an automated test
- December PR-AUC 0.461 (LR) and 0.468 (XGBoost) vs base rate 0.335
- XGBoost only competitive once forced to use very large leaves
- Rolling Platt follows the drift: ECE 0.102 → 0.047
- Rescheduling: earlier is better, but gains within ±2 h are small

Limitations

- No weather forecasts: day-level weather caps accuracy
- One year of NYC departures only
- Rescheduling is associational

Phase 2: deep learning

- MLP with embeddings for carrier, origin and destination
- Same split, features and metrics for a fair comparison

Thank you